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WIEN

Data management plan (DMP)

FAIR Data Science – Coffee Productivity Prediction

FDSP

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1.0	2026-05-26	First version of the DMP – created for the start of the project

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List of acronyms

DMP	data management plan
RDM	research data management
FAIR	Findable, Accessible, Interoperable, Reusable
ML	machine learning
API	Application Programming Interface
CSV	comma-separated values
DOI	digital object identifier
ORCID	Open Researcher and Contributor ID

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Introduction

Science Europe practical guide, FAIR data

A DMP is a structured document that keeps record of what research data is created and what happens to that data during and after a project. It helps with planning the research process and defining responsibilities in a research project involving several researchers or institutions.

For writing this DMP, we followed [the recommendations of Science Europe](#) as they reflect the guidelines agreed upon by the major funders in Europe.

To make our data FAIR, they generally will be treated according to the following criteria:

- We will make our data findable, by uploading it to a data repository that provides a persistent identifier and adding relevant metadata.
- We will make our data accessible by providing open access to data, wherever possible. In cases, where open access is not possible, we will provide meaningful metadata plus contact information for access requests.
- We will make our data interoperable by providing and describing data in a way that is common within our domain by using the same file formats, schemas and vocabularies. We will provide good documentation for all our datasets.
- We will make our data reusable by adding metadata and comprehensive Readme files to all published datasets. The descriptions include details on the methodology used, analytical and procedural information. In case of publication, licenses for code and data will always be assigned and clearly marked.

Relevant Policies and Guidelines

- European Commission's document on Ethics and Data Protection:
https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ethics-and-data-protection_he_en.pdf
- Other (e.g. from a project partner)

1. Data description

1a Lists of datasets that will be reused or produced

Produced datasets

dataset ID	title	type	format	estimated volume	contains sensitive data
P1	Coffee Productivity Dataset	Structured text	CSV, JSON	100 - 1000 MB	no

Description for "Coffee Productivity Dataset":

Dataset containing coffee consumption, sleep, health, lifestyle, and participant related variables used for FAIR data science processing and machine learning experiments. The dataset is transformed into normalized relational tables and enriched with metadata and FAIR documentation during the project.

Reused datasets

1b Data generation and reuse

Methods and software used for data generation and reuse

The project reuses an existing public dataset related to coffee consumption, sleep, lifestyle, and productivity related indicators. The dataset is processed, cleaned, normalized, and transformed into relational tables as part of the FAIR data management workflow. Research data is generated through preprocessing, metadata enrichment, relational schema creation, FAIR metadata generation, machine learning experiments, and derived analytical outputs. Additional artefacts such as DBRepo tables/views, metadata files, model descriptions, and FAIR documentation are also created during the project. The project uses Python based data science workflows implemented in Jupyter Notebooks. Main software and libraries include Python, pandas, scikit-learn, NumPy, matplotlib, DBRepo Python API, JSON based metadata standards, and Git/GitHub for version control and collaboration.

2. Documentation and data quality

2a Data organisation, metadata and documentation

Research data will be structured using normalized relational tables together with processed CSV and JSON representations. The project repository follows a clear folder structure separating raw data, processed data, notebooks, metadata artefacts, documentation, and machine learning outputs. Filenames follow consistent naming conventions and include descriptive identifiers and timestamps where appropriate. Versioning and collaboration are managed using Git and GitHub. Changes to datasets, notebooks, metadata files, and FAIR artefacts are tracked through repository history and incremental updates throughout the project.

This project uses several domain-specific and general metadata standards. For semantic annotation, attributes are mapped to concepts in NCIT (NCI Thesaurus) for health and lifestyle variables, CHEBI for chemical entities (caffeine), and QUDT for units of measurement. General metadata is provided

through README files, dataset descriptions, DBRepo metadata entries, and JSON-LD based metadata artefacts (CodeMeta, RO-Crate, Croissant, FAIR4ML), as well as a Model Card for the trained model.

Additionally, we will provide common metadata such as title, description or keywords when publishing data in open access repositories. In such a case, we will follow the default template provided by the repository, such as Data Cite Metadata or Dublin Core.

As far as possible, we will use controlled vocabularies for our data to allow inter-disciplinary interoperability and machine-actionability.

Documentation will be provided through Jupyter Notebooks, repository README files, schema documentation, metadata artefacts, model documentation, and workflow descriptions. The project repository contains reproducible preprocessing and analysis steps together with dependency information and execution instructions. Additional FAIR documentation, validation outputs, and data management information are maintained throughout the project to support transparency, reproducibility, and future reuse of the data and generated artefacts.

2b Data quality control

The following data quality checks will be done: standardised data capture, data entry validation, peer review of data and representation with controlled vocabularies.

3. Storage and backup during research process

3a Storage and backup facilities

For the duration of the project, storage and backup of data will be ensured by Gökay YILDIRIM (acting as the person responsible for data management and DMP) in cooperation with the system operator. The data will not be stored on the servers of TU Wien.

P1 (Coffee Productivity Dataset) will be stored on GitHub Repository. External storage will be used because the project primarily uses github based version control and repository storage for collaborative development, dataset management, metadata tracking, and reproducible research workflows. Institutional storage services were not required because the project data size is relatively small and collaborative version controlled storage was already integrated into the project workflow.

3b Data security and protection of sensitive data

We pay strict attention to compliance with the relevant institutional and national data protection policies listed in the introduction of this document. At this stage, it is not foreseen to process any sensitive data in the project. If this changes, advice will be sought from the data protection specialist at TU Wien, and the DMP will be updated.

Access to data during research:

dataset ID	selected project members	all other project members	the public
P1	writing	reading only	no access

All incidents will be handled individually by an incident response team that is maintaining the affected service.

4. Legal and ethical requirements

4a Personal data

At this stage, it is not foreseen to process any personal data in the project. If this changes, advice will be sought from the data protection specialist at TU Wien, and the DMP will be updated.

4b Intellectual property rights and rights of use

The following individual(s) hold rights and control access to the project data: Access control over the datasets will be managed by the project team members and repository maintainers. Write access will be limited to authorized project contributors, while read access may be granted to other project members for collaboration and review purposes. The datasets and metadata stored in DBRepo, GitHub repositories, and related FAIR documentation platforms will be managed collectively by the project team throughout the project lifecycle.

4c Ethical issues

No particular ethical issue is foreseen with the data to be used or produced by the project. This section will be updated if issues arise.

5. Data sharing and long-term preservation

5a Data publication and access conditions

As far as possible, obtained datasets will be published in repositories. Details on access conditions, reuse licenses, reasons for restrictions, etc. are collected in the table below.

dataset ID	access conditions	estimated publication date	location for publication (repository)	PID	license
P1	Open	2026-07-15	TU Wien Research Data	DOI	CC-BY-4.0

Repository description:

TU Wien Research Data is an institutional repository of TU Wien to enable storing, sharing and publishing of digital objects, in particular research data. It facilitates the funders' requirements for open access to research data and the FAIR principles by making research output findable, accessible, interoperable, and reusable. A DOI is assigned to each dataset published in TU Wien Research Data. This service is developed by the TU Wien Center for Research Data Management and hosted by TU.it. <https://researchdata.tuwien.ac.at/>

Methods or software needed to access and use data: The published dataset will be provided in common formats such as CSV and JSON, which can be accessed using standard spreadsheet software or programming environments. Reuse of the data and associated workflows may benefit from tools such as Python, Jupyter Notebooks, pandas, NumPy, scikit learn, and DBRepo related tooling. Documentation and README files will be included to support reproducibility and ease of reuse.

5b Long-term preservation and deletion of data

dataset ID	location for long-term storage	minimum retention period (≥ 10 years)	foreseeable research uses and/or users
P1	TU Wien Research Data	10 years	Students, researchers, and the general public interested in data science, machine learning workflows, reproducible research, and FAIR data management practices.

6. RDM responsibilities and resources

6a RDM-roles and responsibilities

The Gökay YILDIRIM will direct the data management process overall, with the research assistants responsible for ensuring metadata production, day-to-day cross-checks, back-up and other quality control activities are maintained.

6b Resources

There are no costs dedicated to data management and ensuring that data will be FAIR.